

PROCESS & BUILD DOCUMENTATION — FOR EXTERNAL AI AUDIT

Home Services Website Studio

Full account of the pipeline built, and the first client site produced with it:
Clean Lab by EZ (mobile auto detailing, Central Florida)

Compiled 2026-07-24 · Repository: Custom Websites (local, git-initialized, single commit at time of writing)
Prepared for import into a third-party AI review tool to critique the studio system and the build for improvement opportunities.

Contents

1. Part 1 — The Studio System

- a. Premise and goal
- b. Repository layout
- c. The six-stage pipeline
- d. The five hard rules
- e. Anti-repetition ledger

2. Part 2 — The Reusable Template (`_template/`)

- a. Stack and exact versions
- b. File inventory
- c. Page inventory
- d. Component inventory
- e. Content collection schema
- f. Design-token architecture
- g. Booking-provider abstraction
- h. Lead-form architecture
- i. Bugs found and fixed at the template level

3. Part 3 — Client Build: Clean Lab by EZ, Stage by Stage

- a. Intake — what is confirmed vs. pending
- b. Stage 0 — Asset audit
- c. Stage 1 — Brand read
- d. Stage 2 — Design directions (three options, one approved)
- e. Stage 3 — Architecture
- f. Stage 4 — Design system / tokens
- g. Stage 5 — Build output
- h. Stage 6 — QA (three parallel audits)

4. Part 4 — Post-QA Iteration (three rounds of direct feedback)

- a. Hero rebuild
- b. Services mega-menu
- c. Hero lead-capture form
- d. Real photo integration
- e. Real logo integration
- f. Real brand-color correction

5. Part 5 — Verified Current State (checked at time of writing)

6. Part 6 — Known Gaps and Open Items

7. Part 7 — Questions for the Auditor

Part 1 — The Studio System

1a. Premise and goal

This repository is a production system for building marketing websites for local home-service and personal-care businesses — plumbing, roofing, general contracting, handyman, windows/glass, in-home sales, mobile detailing, barbershops, salons, pet groomers. The stated premise, verbatim from the project's governing document (`CLAUDE.md`):

"The premise: distinctive on the surface, identical underneath. No two client sites should look like they came from the same shop. Every site converts the same way, because conversion architecture is engineering, not taste."

In practice this means: one shared, hardened Astro codebase (`_template/`) supplies every structural, technical, and compliance guarantee — schema markup, accessibility baselines, lead-capture wiring, SMS/TCPA compliance scaffolding, performance budget — and every client engagement then diverges only in the surface layer: color tokens, type pairing, hero/services/nav "recipe," photography, and copy voice. The goal is to make each client site look like a different design shop produced it, while every site shares the same underlying conversion engineering.

1b. Repository layout

Custom Websites/	
clients/<slug>/	one directory per client engagement
intake/	completed intake form, call notes
assets/	client-supplied photos, logo, reviews, existing site
brief/	00-audit.md → 04-design-system.md (generated, in order, human-approved at s
site/	the actual Astro project for that client
_system/	
RECIPE-LOG.md	anti-repetition ledger — checked before every Stage 2
intake-form.md	the intake form sent to every new client
_template/	the Astro starter, cloned per-client by scripts/new-client.sh
CLAUDE.md	governing rules for this repo (reproduced in 1d below)

1c. The six-stage pipeline

Each client is walked through six numbered stages, each backed by a dedicated Claude Code skill. Stages 1 and 2 are hard-gated behind explicit human approval before the pipeline is allowed to continue.

Stage	Skill	Output	Gate
0 — Audit	/new-client	brief/00-audit.md	none
1 — Brand read	/brand-read	brief/01-brand-read.md	human approval required
2 — Directions	/directions	brief/02-design-directions.md	human approval required (must contain an APPROVED: line)
3 — Architecture	/architect	brief/03-architecture.md	none
4+5 — Build	/build-site	brief/04-design-system.md , populated site/	none
6 — QA	/ship	pass/fail report	none

A PreToolUse hook technically enforces the Stage 2 gate: it blocks any write to `clients/*/site/` until `brief/02-design-directions.md` contains a literal `APPROVED:` line. This is not a social convention — it is mechanically enforced by the harness, independent of model behavior.

1d. The five hard rules (from `CLAUDE.md`, verbatim intent)

- No invented facts.** Every factual claim on a site — years in business, certifications, awards, license numbers, "family owned" — must trace to `intake/`. If a fact isn't there, the field ships blank/flagged rather than guessed.
- No repeated recipes.** `_system/RECIPE-LOG.md` must be read before every Stage 2. A hero archetype or services layout may never repeat within the same vertical (e.g. two mobile-detailing clients can't both get "color-block hero + tier cards"). A row is appended to the ledger after every build.
- Business facts live in `site.config.ts` only.** Phone, hours, license, service areas, NAP (name/address/phone) are never hardcoded into components — changing a client's phone number should be a one-line config edit, not a grep-and-replace across the codebase.
- Image reading is delegated.** A client asset folder can hold 50+ photos; a dedicated `brand-analyst` subagent reads them in an isolated context so only a written summary re-enters the main conversation.
- Compliance ships with the site.** Any form that collects a phone number must carry an unchecked SMS-consent checkbox with full TCPA language, plus a live privacy policy — without both, the client's A2P 10DLC carrier registration is rejected and their GoHighLevel (GHL) campaigns never send.

A sixth rule governs copy specifically: **copy is design material** — written in the client's actual documented voice, never lorem ipsum, never generic "quality service you can trust" filler.

1e. Anti-repetition ledger

The full, current contents of `_system/RECIPE-LOG.md` at time of writing (this is its first-ever entry — Clean Lab by EZ was the system's first client end-to-end):

Date	Client	Vertical	Photo grade	Hero	Services	Nav	Type direction	Signature element
2026-07-24	Clean Lab by EZ	Mobile detailing	C	06 color block	08 package/tier cards	06 left logo/center nav/CTA pill	Technical/performance (Space Grotesk + Inter + JetBrains Mono)	Live vehicle-size price selector (Sedan/SUV/Truck)

Rule encoded in the ledger's header: within a vertical, never repeat the hero archetype or services layout; across the whole roster, never repeat a full hero+services+type combination. If a vertical's option menu starts feeling exhausted, the fix is to extend the menu in the `design-playbook` skill, not to repeat a recipe.

Part 2 — The Reusable Template (`_template/`)

2a. Stack and exact pinned versions

Every dependency is pinned to an exact version (no carets/tildes) and installed for real (`node_modules` present, not just declared):

Package	Version
astro	7.1.3
@astrojs/sitemap	3.7.3
@astrojs/check	0.9.9
@tailwindcss/vite	4.3.3
tailwindcss	4.3.3
typescript	5.7.3
@fontsource/space-grotesk, /inter, /jetbrains-mono	5.3.0

Output mode: `static` . Astro 7's Content Layer API is used for content collections (`src/content.config.ts` , not the legacy `src/content/config.ts` path). Tailwind v4 has no `@astrojs/tailwind` integration package in this major version — it is wired in via the `@tailwindcss/vite` plugin instead; `tailwind.config.mjs` is left as plugin registration only, and the actual design-token layer lives in `src/styles/tokens.css` as a Tailwind v4 `@theme` block.

2b. Design-token architecture — the sentinel-magenta pattern

The template ships with every token *key* a component might reference already present in `tokens.css` , but filled with a loud, impossible-to-miss placeholder value (`#FF00FF` magenta) rather than a plausible-looking default. This means: (1) the template builds and type-checks clean with zero client-specific data, and (2) if a Stage 4 build ever ships without overwriting a token, the mistake is visually screaming rather than quietly shipping a generic gray/blue theme that could pass for intentional. Per-client token values are written once, at Stage 4, into that same file — the generic template file itself is never touched per-client; each client's `site/src/styles/tokens.css` diverges from the template's copy on purpose.

2c. Page inventory (10 pages, template default)

```
/                home
/about/
/contact/
/faq/
/privacy/
/reviews/
/service-areas/  index
/service-areas/[city]/  dynamic, one per areas collection entry
/services/        index
/services/[slug]/    dynamic, one per services collection entry
/services/[service]-[city]/  dynamic city×service combination page
/terms/
/404
```

2d. Component inventory

Path	Purpose
layout/Header.astro	Sticky header, scroll-state background, mobile nav toggle, services mega-menu (island script)
layout/Footer.astro	NAP, hours, legal links, social links — reads entirely from <code>site.config.ts</code>
layout/MobileStickyBar.astro	Persistent bottom call/text bar for small viewports
forms/ContactForm.astro	Full lead form: name, phone, email, service, ZIP, message, SMS consent
forms/QuickQuoteForm.astro	Compact hero variant: name, phone, email, service, SMS consent
booking/BookingEmbed.astro + 5 providers/*	Provider-abstracted booking embed: Calendly, GHL, Google, generic "Other" iframe, and a safe FormPhoneFallback when no real embed URL exists yet
schema/LocalBusinessSchema.astro	JSON-LD, accepts a schema.org subtype (e.g. AutoWash) per business
schema/ServiceSchema.astro , FAQPageSchema.astro , BreadcrumbListSchema.astro , PersonSchema.astro	Additional structured-data types per page kind
ui/ServiceCard.astro , AreaCard.astro , ReviewCard.astro , SectionHeading.astro , CTAButton.astro	Core presentational building blocks, entirely token-driven
ui/Lightbox.astro	Image gallery with lightbox — island
ui/BeforeAfterSlider.astro	Before/after image comparison slider — island, exists in template and every client copy, currently unused on Clean Lab by EZ (no before/after photo pair exists yet)
ui/ChatWidget.astro	Chat widget stub — island, gated fully off by <code>site.config.chat.mode</code>
lib/format.ts	Phone/hours formatting, <code>tel:</code> href helpers
lib/lead-form.ts	Shared submit/fallback logic for every lead-capture form on a site (see 2h)
lib/schema.ts	Shared JSON-LD builder helpers

Per `CLAUDE.md` convention, islands (client-side JS) are restricted to: chat widget, mobile nav, lightbox, before/after slider, booking embed. Everything else renders as static HTML with zero client JS.

2e. Content collection schema (7 collections, Astro Content Layer API)

Defined in `src/content.config.ts`. Six were specified by the studio's `build-standards` skill; a seventh, `cityServicePages`, was added because the city×service combination pages need real authored local prose per combination — a plain data array in `site.config.ts` can't hold freeform text with the swap-test discipline the build requires.

Collection	Key schema fields
services	name, slug, isPrimary, priceFrom?, shortDescription, heroImage?, order
areas	city, state, slug, county?, blurb, neighborhoods?, heroImage?
blog	title, description, publishDate, updatedAt?, heroImage?, draft
team	name, role, photo?, bookingUrl?, bio
testimonials	author, rating, quote, service?, source (google/facebook/yelp/direct), date?, firstParty:boolean — firstParty gates AggregateRating schema eligibility and is never defaulted true
faqs	question, answer, page (sitewide/service/homepage), serviceSlug?
cityServicePages	serviceSlug, citySlug, neighborhoods?, localAngle

2f. Booking-provider abstraction

`site.config.ts`'s `booking.provider` field is a typed union: `'calendly' | 'ghl' | 'google' | 'other' | 'none'`. `'other'` covers providers without a dedicated component (Housecall Pro, ServiceTitan) via a generic iframe. `'none'` is the safe default: rather than ever rendering a broken or empty embed when a client hasn't supplied a real booking URL yet, the site falls back to a visible call/text prompt. This abstraction is what let Clean Lab by EZ ship a fully functional site before its GHL calendar URL exists.

2g. Lead-form architecture

Both lead-capture forms (full `ContactForm` and compact `QuickQuoteForm`) share one submit handler, `src/lib/lead-form.ts`, rather than duplicating fetch/fallback logic:

```

export function initLeadForms(selector = '[data-lead-form]') {
  document.querySelectorAll<HTMLFormElement>(selector).forEach((form) => {
    if (form.dataset.leadFormBound === 'true') return;
    form.dataset.leadFormBound = 'true';
    form.addEventListener('submit', async (e) => {
      e.preventDefault();
      const webhookUrl = form.dataset.webhookUrl;
      const fallbackPhone = form.dataset.fallbackPhone;
      // No webhook configured yet -> show a visible call/text prompt,
      // never silently swallow the submission.
      if (!webhookUrl) { /* ...status message with fallbackPhone... */ return; }
      // Webhook configured -> POST FormData, show success/failure status.
    });
  });
}

```

Every SMS-consent checkbox across both forms is hardcoded `checked={false}` and `required` — the studio's compliance rule (1d, rule 5) is enforced at the component level, not left to per-client copy discipline.

2h. Bugs found and fixed at the template level

During Clean Lab by EZ's Stage 6 QA (three parallel audits — see Part 3h), two defects were discovered that were not specific to that client — they were latent in `_template/` itself and would have shipped to every future client built from it. Both were fixed in the client site and then **backported into** `_template/` so no future client inherits them.

Bug 1 — non-functional form submit. The shared `CTAButton.astro` component originally rendered every instance as an `<a href="#">`. An anchor tag never fires a form's native `submit` event, so any form using `<CTAButton>Send</CTAButton>` as its submit control was completely inert — clicking it did nothing. Fixed by adding a `type: 'link' | 'submit'` prop; when `type="submit"`, the component renders a real `<button type="submit">` instead of an anchor. The current component's own inline comment states the rule explicitly: `"An <a href="#"> never fires a form's submit event, so a form's send action must never use the href variant."`

Bug 2 — no mobile navigation at all. The template's original header had no mobile menu implementation — on small viewports there was simply no way to reach the nav links. Fixed by adding the hamburger-toggle mobile nav pattern now present in `Header.astro` (visible in the component listing above), with a focus-visible-safe toggle button, `aria-expanded` state, and a dedicated `#mobile-nav` panel.

Part 3 — Client Build: Clean Lab by EZ, Stage by Stage

3a. Intake — what is confirmed vs. pending

Source documents: `flyer-1.PNG` (bilingual EN/ES service flyer with areas-served list), `flyer-2.jpg` (pricing flyer), `google-profile.jpg` (Google Business Profile screenshot), plus a follow-up operator Q&A round on 2026-07-23/24. The intake form (`clients/clean-lab-by-ez/intake/intake-form.md`) marks every one of its 61 questions as either a sourced/cited fact, an operator quote, or explicitly **UNANSWERED** — **PENDING OPERATOR**. Nothing is filled with a guess.

Field	Status
Legal/display name	Confirmed: "Clean Lab by EZ" (operator, 2026-07-24)
Phone	Confirmed: (689) 200-6472, SMS-capable
Email	Confirmed: cleanlabbyez@gmail.com
Address	None — confirmed fully mobile, no fixed location
Hours	Confirmed: Mon–Sat 8:00–17:00, closed Sunday
Service area	Confirmed: Kissimmee, Davenport, St. Cloud, Orlando, Poinciana, Haines City (6 cities, flyer-sourced, operator-confirmed as the complete list)
Services & pricing	Confirmed from flyer-2.jpg: Lab Refresh (\$79/\$99/\$119 by vehicle size), Lab Reset ("Most Popular," \$125/\$150/\$175), plus Pet Hair Removal (\$25) and Engine Bay Cleaning (unpriced) add-ons
Reviews	Confirmed aggregate only: 5.0★ / 25 Google reviews. Operator consented to display, but individual review text is not yet in hand — BLOCKING for a real Reviews page
Booking platform	Operator confirmed intent to use GHL calendar, but supplied no embed URL — BLOCKING
Domain	UNANSWERED
License / insurance	UNANSWERED — flagged as possibly N/A for FL mobile detailing, not assumed
Year founded	UNANSWERED
Brand vibe	Operator's own words, verbatim: <code>"performance, dark, be able to provide the best service possible, and dictate my quality"</code>
The one thing every visitor should do	Operator, verbatim: <code>"book an appointment or ask for more information"</code> — treated as a dual-intent CTA requirement, not a single forced action

3b. Stage 0 — Asset audit

Photo grade: C. 13 files supplied, one of which (`image.jpg`) turned out to be a pre-built marketing graphic with baked-in text and a composited stock-feeling vehicle shot rather than a raw photo — excluded, leaving 12 real photos. Both counts sit under the 20-photo floor the studio's asset-quality matrix requires for a B grade. Of the 12, 7 were graded genuinely strong (sharp, real daylight, varied high-value vehicles: Cadillac Escalade IQ, Corvette C8, two Lexus, two Tesla, one F-150); the other 5 were compromised (a headlight-restoration shot cluttered by unrelated parked cars with no before/after pairing; two interior shots with a promotional flyer-prop and, in one case, a real-estate yard sign visible through the windshield).

Design implication drawn from the grade: type-driven, not photo-forward — 12 usable photos cannot carry a home page, about page, reviews page, services index, 4 service pages, and 6 area pages without heavy repetition, so the brand's existing black/orange/white system was designated the primary design language, with the 7 strong photos used as accents rather than structural hero material.

Real-world brand evidence was found to agree across three independent sources — logo, both flyers, and a crew uniform photo all independently show the same black background / vivid orange accent / white text system, which the audit treated as a locked palette rather than a proposal.

People inventory: exactly one of the 13 supplied photos contains a person, shown from behind with no face visible. Zero owner or crew headshots, zero customer-facing photos. The audit's single highest-leverage recommendation: a one-day phone-shoot capturing (1) a face, (2) one true before/after pair on a real job, and (3) 15–20 more exterior shots in the same style as the 7 strong ones — estimated to move the grade from C to a solid B.

3c. Stage 1 — Brand read

Who they are: a lean, owner-operated mobile detailing business covering six Central Florida cities from a Davenport-area base, no shop or storefront, crew of at least two based on uniform photo evidence. Assessed as a business with real operating history and marketing infrastructure (custom logo, two professionally laid-out flyers, a tiered pricing menu, a 5.0★/25-review Google profile) — "not a weekend side hustle," even though the exact founding date is unknown.

Five adjectives: Precision-minded, performance-driven, mobile-first, locally-rooted, direct.

Five adjectives they are NOT: Budget/bargain, corporate/franchise, fussy/exclusive, sterile-despite-the-lab-conceit, slow/laid-back.

Customer psychology: owns a vehicle worth caring about (photo evidence skews toward newer/higher-value cars); arrives in either routine-maintenance mode or "let it go too long, need a reset" mode. Neither is emergency-level urgency; the dominant emotion is impatience with friction, not distress — this customer is comparing tiers and prices, not panicking.

Voice: the brand read explicitly separates the operator's own plain, unpolished speech (*"we're going to be using GHL calendar. It may change, but for now that's what we're going to do"*) from the flyers' more polished taglines ("Lab grade clean for your car," "We make it EZ"), and directs site copy to default to the plainer register — flagging that the flyers may reflect outside design help rather than the operator's natural voice.

Competitive read: the category defaults to stock soap-suds photography, pastel/corporate-blue palettes, vague "eco-friendly" claims, and templated pricing pages. Clean Lab by EZ's genuine (not proposed) points

of difference: the "lab" precision framing itself, real photo evidence of higher-end vehicles, and an already-locked dark/orange palette instead of industry-standard blue-and-white.

3d. Stage 2 — Design directions (three options presented, one approved)

Per the anti-repetition rule, the recipe log was checked first and found empty (first client) — nothing to collide with. All three directions below were required to be type-driven per the photo-grade-C asset matrix.

	A — "Precision Lab" APPROVED	B — "Service Radius"	C — "The Counter-Read"
Thesis	A precision performance lab for vehicles, not a car wash — cold, dark, technical	Lead with "we come to you" as the literal first interaction, mirroring the flyer's own priority order	Deliberate counter-read to the operator's stated vibe — restrained, premium, "we don't wash cars"
Palette	#0B0B0D near-black / #1C1B1A card / #F47920 orange / #F5F3EF warm white / #3A3D42 steel	Identical to A on purpose — proves the differentiation is structural, not chromatic	#16321F deep racing green / #EDE8DD ivory / #B08D57 brushed gold
Type	Space Grotesk display + Inter body + JetBrains Mono numerics	Bolder condensed display, same body/mono logic	High-contrast display serif + neutral grotesque body
Hero archetype	06 — Color block, hexagon texture, one small offset accent image	10 — Map-anchored: stylized map of the 6 served cities	04 — Editorial masthead, restrained image band
Services layout	08 — Package/tier cards (maps directly to the real Lab Refresh/Lab Reset structure)	05 — Tabbed panels (Refresh/Reset/Add-ons)	10 — Index list, type-only, hover reveals image
Nav	06 — Left logo / center nav / right CTA pill	02 — Two-tier: utility strip (phone/hours/"100% Mobile") above main nav	04 — Centered logo, split nav
Signature element	Live interactive Sedan/SUV/Truck price selector	Hover/tap-a-city map reveal with a real local blurb per city	A single restrained "by appointment" concierge-style indicator
Computed AA contrast	Orange-on-near-black 7.12:1, orange-on-card 6.23:1 — both pass AA	Same palette as A, same ratios	Ivory-on-green 10.19:1 (AAA); gold-on-green 4.03:1 — passes only for large text/UI, explicitly restricted from small body copy

The direction document's self-check explicitly notes Direction A was "earned, not reflex" — independently confirmed by the operator's own words, the photo evidence, and the existing brand assets — while B and C exist specifically so the set isn't "the obvious choice in three palettes." Direction C was included deliberately as a genuine counter-read that contradicts the operator's stated vibe, framed as a real option to accept or reject, not a recommendation.

Result: Direction A, "Precision Lab," approved with no modifications.

3e. Stage 3 — Architecture

Modeling decision: the `services` collection was built to match exactly what the business sells — two real tiered packages (Lab Refresh, Lab Reset) plus two priced add-ons (Pet Hair Removal, Engine Bay Cleaning) — rather than inventing generic categories like "Interior Detailing" / "Exterior Detailing" that don't correspond to how the business actually prices or sells. The architecture doc flagged, in advance, that Pet Hair Removal and Engine Bay Cleaning each had only one flyer line of description and zero supporting photos, making the studio's 600+-word service-page floor a real risk for those two pages specifically — flagged before Stage 5, not discovered at Stage 6 (see Part 5 for the current actual word counts).

Page map: 17 static/dynamic URL patterns defined up front (home, services index + 4 service pages, service-areas index + 6 area pages, about, reviews, contact, faq, privacy, terms, 404), each with its own title tag, H1, and required schema type pre-specified.

City × service matrix: deliberately capped at 12 (2 primary services × 6 cities), explicitly *not* extended to the add-on services, because writing "genuinely local" copy for services that don't yet have enough source material would fail the studio's own swap-test discipline. Each of the 12 combinations was pre-assigned a real, verifiable regional fact (public geography/climate/seasonal knowledge — e.g. Davenport's HOA-community driveway access near Posner Park/Championsgate, Kissimmee's tourist-corridor rental-vehicle turnover, Haines City's citrus-grove-adjacent rural approach roads) for the copywriter to develop at Stage 5, explicitly distinguished from a claim about the business itself. A genuine, well-known Central Florida detailing angle — seasonal "love bug" swarms, a real regional paint/grille problem worst in late spring/early fall — was flagged as available but not yet used anywhere in copy.

/es/ Spanish site: intentionally not built. `flyer-1.PNG` is entirely in Spanish, a real signal of Spanish-speaking customer reach, but the intake question asking whether a full Spanish site is wanted was left unanswered — flagged as a real opportunity rather than assumed.

Internal linking graph (not a flat sitemap dump): every city×service page links back to both its parent service page and its parent area page; both services and service-areas roll up to their index pages, which roll up to home.

Two items explicitly flagged as launch-blocking in this document (repeated from intake/audit because they land directly in `site.config.ts`): the GHL booking embed URL, and the GHL forms webhook URL. Both ship with safe fallbacks (`booking.provider: 'none'`; forms degrade to a visible call/text notice) rather than a broken embed, but neither is genuinely launch-ready until real URLs exist.

3f. Stage 4 — Design system / tokens

This is the point at which `site/` begins diverging from `_template/` on purpose — the sentinel-magenta placeholders in `tokens.css` are overwritten with real, computed values specific to Clean Lab by EZ. The original Stage 4 values below were later corrected twice more (once by Stage 6 QA, once by direct pixel-sampling of the real logo — see Part 4f); the table shows the **final, current** values as they exist in the repository today.

Token	Value	Use	Computed AA contrast
<code>--color-surface</code>	 #0B0B0D	Page background	—
<code>--color-surface-raised</code>	 #1C1B1A	Card/panel background	—
<code>--color-brand-primary</code>	 #F78C16	Single electric accent — CTAs, links, price numerics	8.16:1 on surface, 7.14:1 on surface-raised
<code>--color-brand-secondary</code>	 #C06D11	Hover/pressed states	decorative
<code>--color-brand-accent</code>	 #FFC34D	"Most Popular" badge glow only	decorative
<code>--color-text-primary</code>	 #FFFFFF	Body copy, headings	19.66:1 on surface, 17.2:1 on surface-raised
<code>--color-text-muted</code>	 #B8B4AC	Captions, utility labels	9.52:1 on surface
<code>--color-border</code>	 #666B74	Functional UI boundaries — form inputs, price-selector tabs	3.67:1 on surface, 3.21:1 on surface-raised
<code>--color-divider</code>	 #2A2C30	Decorative-only section dividers	not relied on for a11y

Naming decision: because this brand is dark-first (near-black base, light text), the token names were deliberately written as explicit semantic pairs (`--color-text-primary` / `--color-text-inverse`) instead of forcing them into the usual numbered `neutral-50...900` scale, which would have had to lie about which end of the scale is "dark."

Correction made at Stage 4 itself, before Stage 6: the Stage 2 sketch had estimated the border/divider gray at `#3A3D42` . Computed against the real `#0B0B0D` surface, that value only cleared 1.80:1 — well under WCAG's 3:1 minimum for perceivable UI boundaries. The color actually used for functional borders was lightened to a value that passed; the original darker tone was kept only for purely decorative dividers that carry no information if invisible.

Type system: self-hosted via `@fontsource` (no external font requests, no CDN layout shift) — Space Grotesk (700/500) for display, Inter (400/500) for body, JetBrains Mono (500) reserved specifically for every price and spec number on the site. A 9-step type scale runs from 14px utility text to 56px hero display, plus two dedicated monospace sizes for the price selector's live numbers.

Motion/shadow tokens: a standard 200ms / fast 120ms duration pair on a shared cubic-bezier ease; a two-layer card shadow (depth + hairline inner highlight, since a flat shadow barely reads against an already-near-black page); an "elevated" shadow variant for hover/floating states.

Rule stated explicitly in this document, enforced from Stage 4 forward: no raw hex value may appear in any component file — every color reference must use one of the named tokens.

3g. Stage 5 — Build output

32 pages built (verified against the current `dist/` output — see Part 5 for the exact enumeration and re-verification). Content authored per the Stage 3 plan: 4 service entries, 6 area entries, 12 city-service-page entries, 22 FAQ entries. Zero testimonial entries were authored — per the Stage 3 plan, individual review text was not yet in hand, and the studio's no-fabrication rule (1d, rule 1) meant the Reviews page ships showing only the confirmed aggregate (5.0★/25 reviews) rather than any invented quotes.

3h. Stage 6 — QA (three parallel audits)

Three specialized read-only audit agents were run against the built site: **SEO** (schema markup, meta tags, NAP consistency, internal linking), **accessibility** (WCAG 2.1 AA — contrast, keyboard focus, semantics, form labels, motion), and **design-critic** (templated defaults, recipe-log collisions, fidelity to the approved Direction A brief). This pass surfaced real, concrete defects rather than passing cleanly:

- The two template-level bugs described in Part 2h (non-functional submit button, missing mobile nav) — found here, fixed in the client, then backported to `_template/`.
- The `--color-border` contrast failure described in Part 3f (2.76:1 against `surface-raised`, where it's actually used most, versus only being validated against `surface` originally) — caught by the accessibility audit specifically, not assumed safe by inspection.
- The `PriceSelector` component's ARIA implementation was caught incomplete and brought up to the full WAI-ARIA APG Tabs pattern — roving `tabindex`, `arrow-key/ Home / End` navigation, `aria-controls / aria-selected` wiring (visible in the component's current source, Part 4 excerpt below; the component's own header comment states this was "caught incomplete at Stage 6 QA").

Part 4 — Post-QA Iteration (three rounds of direct user feedback)

After Stage 6 passed, the user gave three further rounds of direct visual/UX feedback that drove a second wave of real changes, documented here because they represent deliberate design decisions beyond the six-stage pipeline's original scope — the kind of judgment calls an external reviewer should be able to see and question.

4a. Hero rebuild

The original Direction-A hero was a pure color-block (per the approved brief — no full-bleed photo, because the C-grade photo library "can't carry one"). Operator feedback: the color-block-only hero read as plain. The hero was rebuilt to a full-bleed real photo (the strongest single asset from the Stage 0 audit, the black Cadillac Escalade IQ) with a two-layer scrim — a horizontal gradient for left-to-right text legibility plus a vertical bottom gradient — layered under the existing hexagon texture at 60% opacity. The component's own code comment documents the reasoning: `"a single flat gradient reads muddy against a real photo."` One hero word ("driveway.") carries a subtle animated gradient "shine" sweep (`.text-shine` , a 4.5s looping background-position animation, disabled entirely under `prefers-reduced-motion`). Above-the-fold hero content uses a single one-time fade-up-in entrance animation (`.hero-enter` , 700ms, staggered 150ms on the second block) — explicitly *not* a scroll-triggered reveal repeated on every section, which the studio's own `design-playbook` skill bans as a generic-AI-site tell.

4b. Services mega-menu

The header's "Services" nav item was upgraded from a plain link to a real hover/click mega-menu panel showing all 4 services with a thumbnail image and short description each, built from the same `lib/photos.ts` registry used elsewhere. Implementation (see `Header.astro` excerpt) supports click-toggle (for touch), hover-intent with a 150ms close delay (to tolerate the cursor briefly leaving the trigger), `Escape` -to-close, and outside-click-to-close, with full `aria-expanded / aria-haspopup / role="menu"` semantics.

4c. Hero lead-capture form

A compact `QuickQuoteForm` (name/phone/email/service + SMS consent) was added directly into the hero, next to the headline, rather than leaving the hero with only a CTA button — per operator feedback that the hero should lead with an actual quote request. The vehicle-size price selector (the Direction A signature element) was relocated out of the hero into the services section beneath it, so the hero's real estate is spent on lead capture rather than the price-comparison tool.

4d. Real photo integration

A central registry, `src/lib/photos.ts` , was built holding exactly the 8 real Stage-0-audit-approved "strong" photos, each imported once and exposed via a typed key with a pre-written accessible `alt` text. Every page that wants a real photo — service pages, area pages, city-service pages, the header mega-menu, the

homepage gallery — looks the photo up by key through this registry rather than re-importing images ad hoc, and the module's own comment states the discipline explicitly: with only 8 real photos existing, pages are expected to *reuse* them across combinations rather than pretend a unique shot exists for every service/city pairing.

4e. Real logo integration

The user supplied transparent PNGs (icon mark alone, wordmark, "by EZ" script, full lockup) plus two SVG files. The PNGs were used, precisely re-cropped with Python/PIL. Of the two SVGs: `logo3.svg` was evaluated and rejected — an 11,749-path auto-traced export, unusable bloat for a production asset — while `logo.svg` (a genuine 324-path hand-built vector) was kept in the asset folder as available but is not currently wired into any component (see Part 6). The header and footer now render the real brand lockup: the icon mark, "Clean" in white, "Lab" in the brand orange (matching the real logo's two-tone treatment), and the actual "by EZ" script asset — verified directly in both `Header.astro` and `Footer.astro`, which implement the identical lockup pattern independently in each file rather than sharing a subcomponent.

4f. Real brand-color correction via pixel sampling

The Stage 2/4 orange (`#F47920`) had been estimated from the flyer artwork. Once the real logo PNG files were available, the actual brand orange was pixel-sampled directly from the source files rather than re-estimated visually: **`#F78C16`**, not `#F47920` — the icon's accent color and the "by EZ" script color were found to agree exactly. The body-text color was corrected the same way: pure white (`#FFFFFF`), not the warm cream (`#F5F3EF`) originally approximated. Every AA contrast ratio in the token table (Part 3f) was recomputed against the corrected values before the swap was made — all passed, and several improved versus the original estimate (e.g. the primary orange went from 7.12:1 to 8.16:1 against the page background).

Part 5 — Verified Current State

The following was re-checked directly against the repository at the time this document was compiled (2026-07-24), rather than taken on trust from prior session notes, per this system's own working practice of verifying claims against current file/build state before restating them.

Check	Result
Git status	Single commit exists on <code>main</code> (<code>4427585 chore: studio scaffold</code>). Every file described in this document — the entire <code>_template/</code> build and the entire <code>clients/clean-lab-by-ez/</code> client — is uncommitted working-directory state . Nothing described here has been pushed or persisted to git history.
<code>astro check</code> , client site	Re-run just now: 0 errors, 0 warnings, 56 hints across 50 files. The 56 hints are all the same informational notice — the JSON-LD <code><script type="application/ld+json" set:html=...></code> pattern in the four schema components triggers an Astro hint recommending an explicit <code>is:inline</code> directive. Cosmetic; not an error, not previously mentioned in prior session notes.
Built page count, client site	32 pages, counted directly from <code>dist/</code> : <code>home</code> , <code>404</code> , <code>about</code> , <code>contact</code> , <code>faq</code> , <code>privacy</code> , <code>reviews</code> , <code>terms</code> , <code>services index</code> + 4 service pages + 12 city-service pages, <code>service-areas index</code> + 6 area pages.
Built page count, template	10 pages (the generic set, no dynamic collection entries).
Service-page word counts vs. the studio's own 600-word floor	GAP — not previously flagged as resolved — see callout below.

New finding, surfaced while compiling this document: the studio's build-standards skill sets an explicit floor for service detail pages — "600+ words, own FAQ, own CTA. Not a stub." The four live service pages, measured just now by direct word count:

Page	Word count	Vs. 600-word floor
lab-refresh.md	529	71 words short
lab-reset.md	500	100 words short
pet-hair-removal.md	211	389 words short
engine-bay-cleaning.md	173	427 words short

This matches the Stage 3 architecture document's own advance warning (Part 3e) that pet-hair-removal and engine-bay-cleaning specifically were a "real Stage 5 challenge" given one flyer line and zero photos each — but the gap was never subsequently closed, and no QA pass flagged the two primary tier pages (lab-refresh, lab-reset) also landing under the floor. This was not caught by the Stage 6 SEO/a11y/design-critic audits described in Part 3h, and is worth this document's audit AI weighing in on: whether to write more real content (there is a genuine word-count ceiling on what can be said honestly, per Rule 1 — no invented facts), reduce the stated floor for add-on-tier pages specifically, or treat this as acceptable for a pre-launch build.

Part 6 — Known Gaps and Open Items

Item	Status
Nothing in this session's work is committed to git	ACTION NEEDED — a deliberate choice to defer, not an oversight, but worth resolving before more work stacks on top of an uncommitted tree
GHL booking embed URL	PENDING — blocking real booking; safe fallback shipped in the meantime
GHL forms webhook URL	PENDING — blocking real lead capture; safe fallback shipped in the meantime
License/insurance info	PENDING — may be genuinely N/A for FL mobile detailing, needs a direct confirm either way
Individual review text	PENDING — only the real aggregate rating is shown; no fabricated quotes exist
Real domain	Placeholder <code>cleanlabbyez.com</code> in <code>astro.config.mjs</code> , explicitly commented as unconfirmed
<code>logo.svg</code> (real 324-path vector)	Present in assets, not wired into any component — PNGs are used instead
<code>BeforeAfterSlider.astro</code>	Built in both <code>_template/</code> and the client site, currently unused — no real before/after photo pair exists yet (the Stage 0 audit's top recommendation)
Chat widget	Component exists, <code>site.config.chat.mode: 'off'</code> — intentionally disabled, future scope
Service-page word counts	See Part 5 callout — 4 of 4 pages under the studio's own 600-word floor
Git author identity	Still auto-detected from the OS account, never explicitly configured for this repo
<code>/es/</code> Spanish site	Flagged as a real opportunity (flyer-1 is fully Spanish) but not built — intake question unanswered

Part 7 — Questions for the Auditor

This document is being handed to a separate AI system for critique. Areas where a second opinion would be genuinely useful:

1. Is the studio's own no-fabrication rule (never invent a fact not in intake) in tension with its 600-word service-page floor, when a client's real source material is this thin? Which should give — write thinner, honest pages, or push harder for real source material before Stage 5?
2. Is Direction A ("Precision Lab") — fully dark, single-orange-accent, tier-card services — actually the strongest choice among the three presented, or does the self-check in Part 3d underrate Direction C's counter-read given how saturated dark/orange is becoming across other verticals in this same studio's roster?
3. Is the recipe-log granularity (hero archetype + services layout + nav + type direction, logged per client) fine enough to actually prevent two future clients in the same vertical from feeling similar, or does it need a finer axis (e.g. photo treatment, motion signature)?
4. Does the booking-provider abstraction (`calendly` | `ghl` | `google` | `other` | `none`) and the shared `lead-form.ts` fallback pattern generalize cleanly, or are there provider-specific edge cases (e.g. GHL's actual webhook payload shape) this design hasn't been tested against yet?
5. Any structural or accessibility issues visible in the code excerpts in Parts 2 and 4 that the studio's own three Stage-6 audits would plausibly miss again on the next client?

End of document. Compiled directly from the repository at `/Users/marcelbrun/My Documents/Claude Code Projects/Custom Websites` — brief documents in `clients/clean-lab-by-ez/brief/`, intake in `clients/clean-lab-by-ez/intake/intake-form.md`, recipe ledger in `_system/RECIPE-LOG.md`, and source files under `_template/` and `clients/clean-lab-by-ez/site/`, all read directly rather than summarized from memory.